



TMDB Movie Pipeline

End-to-End Data Pipeline

Bronze → Silver → Gold → Dashboard

By : Nabila Hulwana Z.

Final Project Bootcamp Data Engineering rubythelib.ai

Project in github : [CLICK HERE](#)

Overview : Studi Kasus & Dataset

Project Context

Membangun pipeline data end-to-end untuk menganalisis industri film global menggunakan Medallion Architecture

Key Questions

1. Bagaimana tren revenue/ keuntungan industri film dari tahun 1992-2016?
2. Genre mana yang paling profitable secara konsisten?
3. Sutradara mana dengan ROI tertinggi sepanjang karir?
4. Film mana saja yang masuk kategori Mega Blockbuster (performa sangat tinggi)?

Data Description (Secondary-dataset)

1 TMDb API

- Tipe Data : JSON
- Link : [klik ini](#)
- 80 rows
- API Key, Endpoint/ movie/ popular

2 Kaggle – tmdb_5000_movies.csv

- Tipe Data : CSV
- Link : [klik ini](#)
- 4673 rows
- Budget, revenue, genres, keywords, companies

3 Kaggle – tmdb_5000_credits.csv

- Tipe Data : CSV
- Link : [klik ini](#)
- 4801 rows
- Cast dan crew per film

Tools

DATABASE



ClickHouse

INFRASTRUCTURE



STREAMING



TRANSFORM



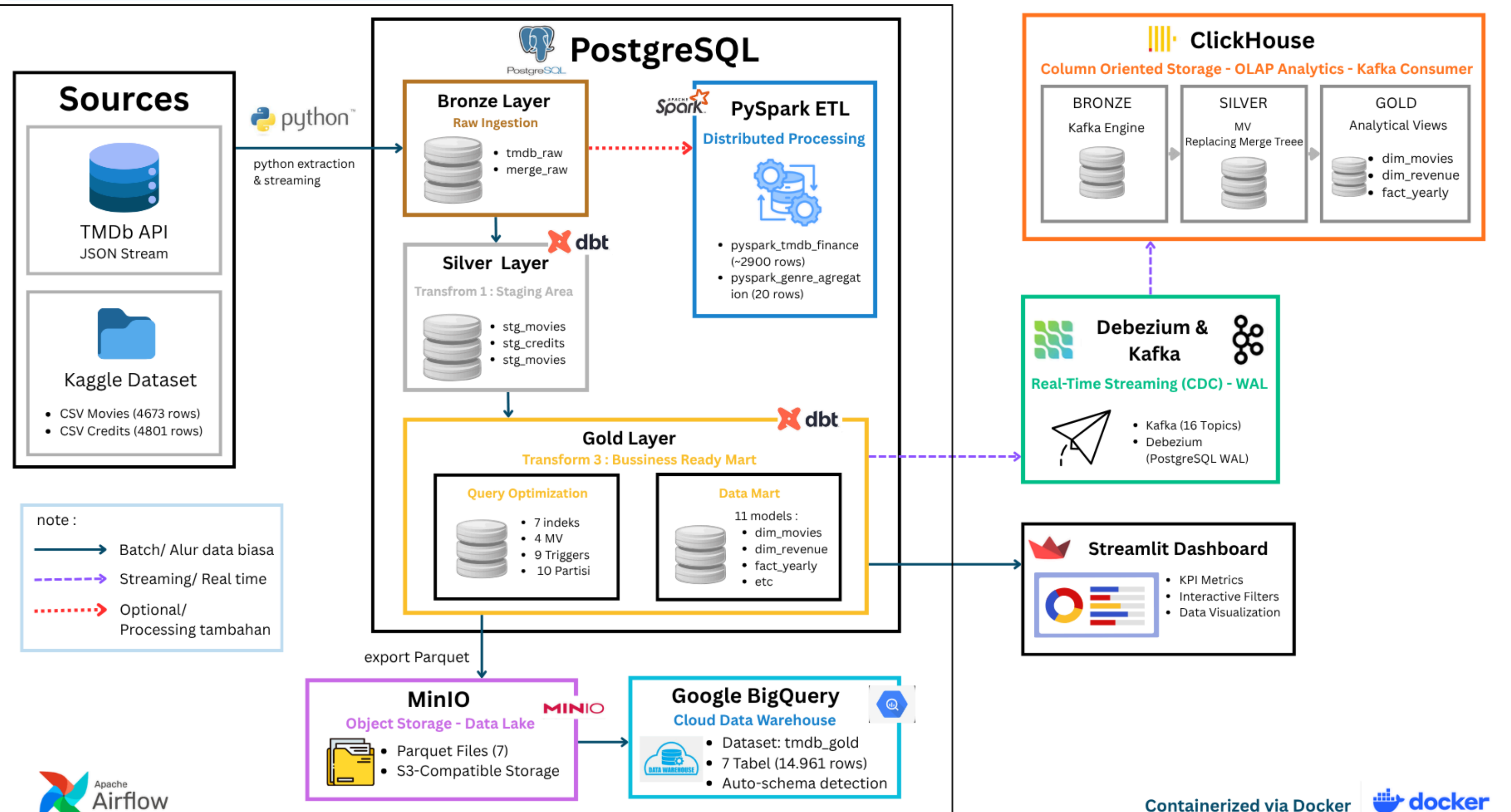
PROCESSING



STORAGE



End-To-End Data Engineering Pipeline : TMDB MOVIE



Containerized via Docker



Step 1: Bronze Layer

Why?

Data raw disimpan tanpa transformasi apapun.

Docker-Setup

1. postgres:13 (Docker)
2. dpage/pgadmin4 (Docker UI)

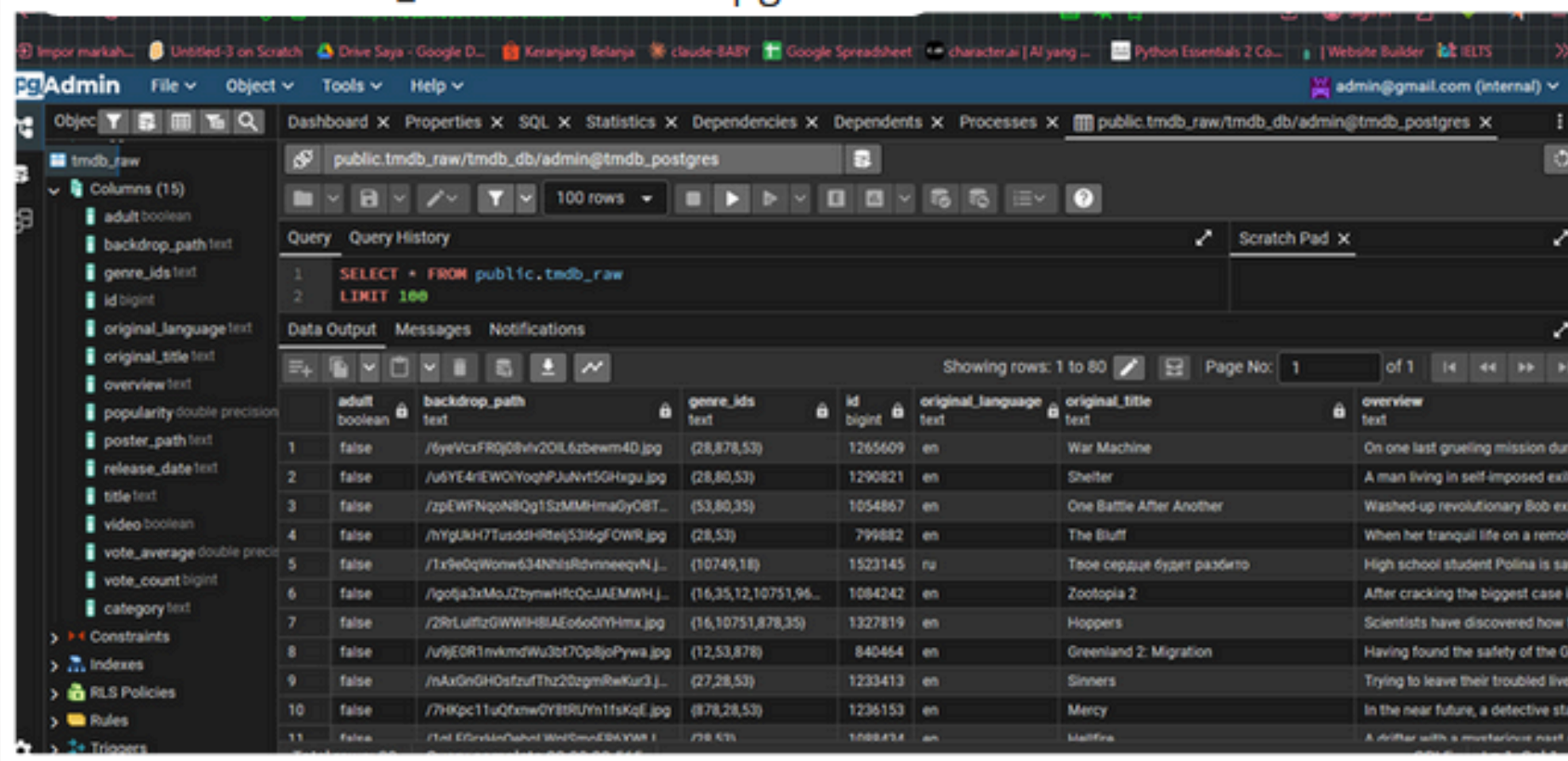
Data-Sources

1. TMDb API (80 rows)
2. credits raw csv file from Kaggle (4.803 rows)
3. movies raw csv file from Kaggle (4.807)

Output-Tables

1. **tmdb_raw** (15 Kolom, 80 rows)
2. **merge_raw**, gabungan 2 file csv Kaggle (22 kolom, 4.803 baris)

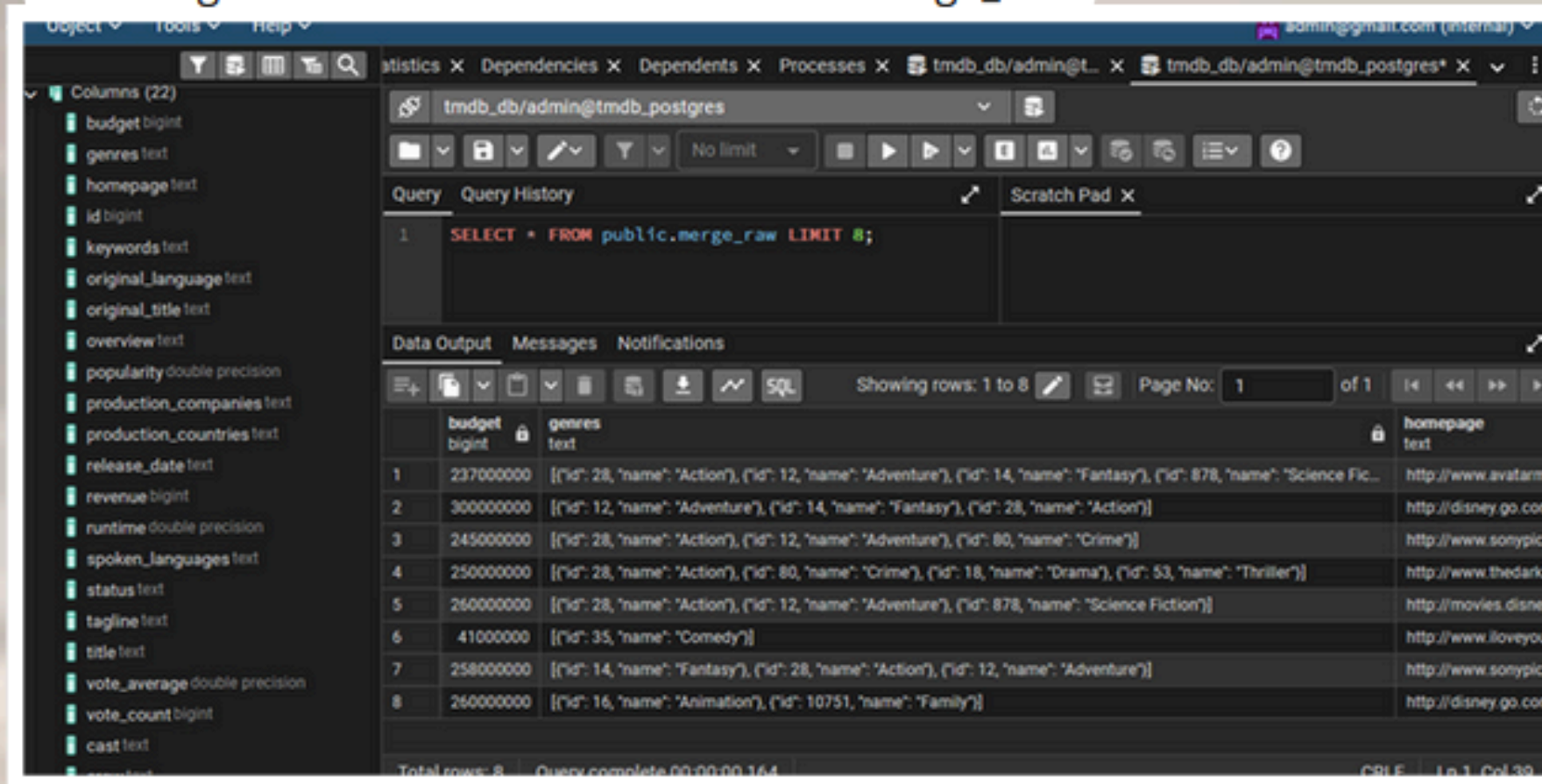
Load tabel tmdb_raw dari API ke pgAdmin



The screenshot shows the pgAdmin interface with the 'tmdb_raw' table selected. The table has 15 columns: adult, backdrop_path, genre_ids, id, original_language, original_title, overview, popularity, poster_path, release_date, title, video, vote_average, vote_count, and category. The data is displayed in a table view showing rows 1 to 80.

	adult	backdrop_path	genre_ids	id	original_language	original_title	overview
1	false	/6yeVcxFRQj08vhr20IL6zbwm40.jpg	(28,878,53)	1265609	en	War Machine	On one last grueling mission dur
2	false	/u6YE4nEWOYyoghPJaNy15GHxgu.jpg	(28,80,53)	1290821	en	Shelter	A man living in self-imposed exil
3	false	/zpEWFNqoN8Qg1SdMMHma0y0BT...	(53,80,35)	1054867	en	One Battle After Another	Washed-up revolutionary Bob ex
4	false	/hYgUkH7TusddH8teJ53l6gFOWR.jpg	(28,53)	799882	en	The Bluff	When her tranquil life on a remot
5	false	/1x9e0qWonw634NhisRdvneeqvN...	(10749,18)	1523145	ru	Трое cepгue бьют пасторо	High school student Polina is sa
6	false	/lgoja3xMoJZbYmHfcQcJAEMWHJ...	(16,35,12,10751,96...	1084242	en	Zootopia 2	After cracking the biggest case i
7	false	/2RtLuffizGWWiH8IAEo6o0YHmx.jpg	(16,10751,878,35)	1327819	en	Hoppers	Scientists have discovered how t
8	false	/u9E0R1nvkmdWu3t7OpIjoPywa.jpg	(12,53,878)	840464	en	Greenland 2: Migration	Having found the safety of the G
9	false	/nAuGnGH0sfzUTh20zgmRwKu3J...	(27,28,53)	1233413	en	Sinners	Trying to leave their troubled liv
10	false	/7Hkpc11uQfkw0Y8BRUYN1sKqE.jpg	(878,28,53)	1236153	en	Mercy	In the near future, a detective sta
11	false	/tadE0v4Np4ndW4P0vW4E0AYW1...	(758,53)	1088474	en	Matfina	A drifter with a mysterious past

Hasil ingestion dari 2 file CSV ke tabel merge_raw



The screenshot shows the pgAdmin interface with the 'merge_raw' table selected. The table has 22 columns: budget, genres, homepage, id, keywords, original_language, original_title, overview, popularity, production_companies, production_countries, release_date, revenue, runtime, spoken_languages, status, tagline, title, vote_average, vote_count, and cast. The data is displayed in a table view showing rows 1 to 8.

	budget	genres	homepage
1	237000000	['id': 28, 'name': 'Action'], ['id': 12, 'name': 'Adventure'], ['id': 14, 'name': 'Fantasy'], ['id': 878, 'name': 'Science Fic...	http://www.avatarm
2	300000000	['id': 12, 'name': 'Adventure'], ['id': 14, 'name': 'Fantasy'], ['id': 28, 'name': 'Action']	http://disney.go.com
3	245000000	['id': 28, 'name': 'Action'], ['id': 12, 'name': 'Adventure'], ['id': 80, 'name': 'Crime']	http://www.sonypic
4	250000000	['id': 28, 'name': 'Action'], ['id': 80, 'name': 'Crime'], ['id': 18, 'name': 'Drama'], ['id': 53, 'name': 'Thriller']	http://www.thedark
5	260000000	['id': 28, 'name': 'Action'], ['id': 12, 'name': 'Adventure'], ['id': 878, 'name': 'Science Fiction']	http://movies.disne
6	41000000	['id': 35, 'name': 'Comedy']	http://www.loveyou
7	258000000	['id': 14, 'name': 'Fantasy'], ['id': 28, 'name': 'Action'], ['id': 12, 'name': 'Adventure']	http://www.sonypic
8	260000000	['id': 16, 'name': 'Animation'], ['id': 10751, 'name': 'Family']	http://disney.go.com

Step 2 : Silver Layer

Purpose

Membersihkan data sebelum ke proses analitik

3 Staging Models :

stg_movies.sql

sumber : merge_raw

- 1. Handle 2 format tanggal
- 2. parse kolom berisi JSON
- 3. Normalisasi popularity & vote_avg

stg_credits.sql

sumber : merge_raw

Ekstrak lead actor, top 3 cast, director, screenplay writer, producer, cinematographer, music composer dari kolom JSON

stg_tmdbapi.sql

sumber : tmdb_raw

- 1. Fix format tanggal
- 2. handle genre_ids sebagai PostgreSQL native array

stg_credits.sql

	movie_id bigint	title text	top_3_cast text	lead_actor text	total_cast_count integer
1	19995	Avatar	Sam Worthington, Zoe Saldana, Sigourney Weaver	Sam Worthington	
2	285	Pirates of the Caribbean: At World's End	Johnny Depp, Orlando Bloom, Keira Knightley	Johnny Depp	
3	206647	Spectre	Daniel Craig, Christoph Waltz, Léa Seydoux	Daniel Craig	
4	49026	The Dark Knight Rises	Christian Bale, Michael Caine, Gary Oldman	Christian Bale	
5	49529	John Carter	Taylor Kitsch, Lynn Collins, Samantha Morton	Taylor Kitsch	
6	16538	I Love You, Man	Paul Rudd, Jason Segel, Rashida Jones	Paul Rudd	
7	559	Spider-Man 3	Tobey Maguire, Kirsten Dunst, James Franco	Tobey Maguire	
8	38757	Tangled	Zachary Levi, Mandy Moore, Donna Murphy	Zachary Levi	
9	99861	Avengers: Age of Ultron	Robert Downey Jr., Chris Hemsworth, Mark Ruffalo	Robert Downey Jr.	
10	767	Harry Potter and the Half-Blood Prince	Daniel Radcliffe, Rupert Grint, Emma Watson	Daniel Radcliffe	
11	209112	Batman v Superman: Dawn of Justice	Ben Affleck, Henry Cavill, Gal Gadot	Ben Affleck	
12	1452	Superman Returns	Brandon Routh, Kevin Spacey, Kate Bosworth	Brandon Routh	
13	10764	Quantum of Solace	Daniel Craig, Olga Kurylenko, Mathieu Amalric	Daniel Craig	
14	58	Pirates of the Caribbean: Dead Man's Chest	Johnny Depp, Orlando Bloom, Keira Knightley	Johnny Depp	

stg_movies.sql

	movie_id bigint	title text	original_language text	overview text
1	19995	Avatar	en	In the 22nd century, a paraplegic Marine is dispatched to the moon Pandora on a un...
2	285	Pirates of the Caribbean: At World's End	en	Captain Barbossa, long believed to be dead, has come back to life and is headed to...
3	206647	Spectre	en	A cryptic message from Bond's past sends him on a trail to uncover a sinister organ...
4	49026	The Dark Knight Rises	en	Following the death of District Attorney Harvey Dent, Batman assumes responsibilit...
5	49529	John Carter	en	John Carter is a war-weary, former military captain who's inexplicably transported to...
6	16538	I Love You, Man	en	Peter Klaven is a successful real estate agent who, upon getting engaged to the wo...
7	559	Spider-Man 3	en	The seemingly invincible Spider-Man goes up against an all-new crop of villain - inc...
8	38757	Tangled	en	When the kingdom's most wanted-and most charming-bandit Flynn Rider hides out...
9	99861	Avengers: Age of Ultron	en	When Tony Stark tries to jumpstart a dormant peacekeeping program, things go aw...
10	767	Harry Potter and the Half-Blood Prince	en	As Harry begins his sixth year at Hogwarts, he discovers an old book marked as 'Pr...
11	209112	Batman v Superman: Dawn of Justice	en	Fearing the actions of a god-like Super Hero left unchecked, Gotham City's own for...
12	1452	Superman Returns	en	Superman returns to discover his 5-year absence has allowed Lex Luthor to walk fre...
13	10764	Quantum of Solace	en	Quantum of Solace continues the adventures of James Bond after Casino Royale. B...
14	58	Pirates of the Caribbean: Dead Man's Chest	en	Captain Jack Sparrow works his way out of a blood debt with the ghostly Davay...

Step 2 : Gold Layer – Schema : silver_mart

Purpose

Data siap sipakai untuk analitik dan dashboard.

Kategori Performa dari tabel dim_revenue

Label	Keterangan
Mega Blockbuster	Film sukses besar, balik modal ≥ 3x lipat
Blockbuster	Film sukses, balik modal ≥ 2x lipat
Profitable	Untung tapi tidak signifikan
Break Even	Tidak untung tidak rugi
Loss	Film merugi

	table_schema name	table_name name
1	public	merge_raw
2	public	tmdb_raw
3	silver_mart	dim_actor
4	silver_mart	dim_company
5	silver_mart	dim_country
6	silver_mart	dim_directors
7	silver_mart	dim_genres
8	silver_mart	dim_language
9	silver_mart	dim_movies
10	silver_mart	dim_revenue
11	silver_mart	dim_yearlysum
12	silver_mart	fact_vs
13	silver_mart	fact_yearly

11 Mart Models

1. **dim_movies**
Tabel master film lengkap, gabungan 3 sumber

2. **dim_genres**
Ringkasan performa per genre

3. **dim_directors**
Profil dan performa karir sutradara

4. **dim_actor**
Profil karir aktor lead dan aktor supporting

5. **dim_country**
Rangkuman film (revenue, rating, dll) per negara

6. **dim_language**
analisis film per bahasa
7. **dim_company**
Profil dan analisis rumah produksi film

8. **dim_revenue**
Ringkasan keuangan per film (ROI, budget, profit, dll)

9. **fact_vs**
Rekonsiliasi 2 sumber data (CSV vs API)

10. **fact_yearly**
Tren industri pertahun dan pergenre.

11. **dim_yearlysum**
Rignkasan film dikelompokkan berdasarkan tahun nya.

Step 3 : Query Optimization

Step by Step :

1 EXPLAIN – Analisis Performa Query

dijalankan pada **dim_movies** dan **dim_revenue** sebagai baseline sebelum optimasi. Lalu **membuat 7 indexes** :

	QUERY PLAN
	text
1	Sort (cost=13.24..13.31 rows=27 width=36)
2	Sort Key: release_year
3	-> HashAggregate (cost=12.26..12.60 rows=27 width=36)
4	Group Key: release_year
5	-> Seq Scan on fact_yearly (cost=0.00..10.45 rows=363 w...
6	Filter: ((release_year >= 1990) AND (release_year <= 20...

Membuktikan indeks bekerja, COST turun drastic

	QUERY PLAN
	text
1	Limit (cost=79.52..79.53 rows=1 width=31)
2	-> Sort (cost=79.52..79.53 rows=1 width=31)
3	Sort Key: revenue DESC
4	-> Bitmap Heap Scan on dim_revenue (cost=8.69..79.51 rows=1 width=31)
5	Recheck Cond: (primary_genre = 'Action'::text)
6	Filter: (vote_average > 7.0)
7	-> Bitmap Index Scan on idx_dim_revenue_genre (cost=0.00..8.69 rows=588 widt...
8	Index Cond: (primary_genre = 'Action'::text)

2 PARTISI - Tabel fact_yearly_partisi menjadi per Dekade.

	decade	total_films	total_revenue	avg_roi_pct	revenue_growth_pct
	integer	bigint	numeric	numeric	numeric
1	1990	540	60946155147	867.37	[null]
2	2000	1335	155003313780	4541.52	154.3
3	2010	947	148615359698	122918.97	-4.1

Step 3 : Query Optimization

3 Materialized View + Auto Refresh Trigger

dibuat untuk cache agregasi untuk dashboard, secara otomatis men-trigger refresh MV tanpa manual.

MV 1 : Performa per negara : negara yang dominan per genre/ profit, genre favorit per negara

The screenshot shows a database interface with a query editor and a results table. The query is `SELECT * FROM public.mv_country_performance`. The results table has 10 columns: `primary_country` (text), `total_films` (bigint), `total_revenue` (numeric), `total_profit` (numeric), `avg_roi_pct` (numeric), `avg_rating` (numeric), `most_common_genre` (text), and `top_grossing_film` (text). The table displays 3 rows of data.

	primary_country	total_films	total_revenue	total_profit	avg_roi_pct	avg_rating	most_common_genre	top_grossing_film
1	United States of Ameri...	3102	280520928071	189414222081	207.90	6.09	Comedy	Avatar
2	United Kingdom	374	36541235127	25848003976	241.72	6.46	Drama	Skyfall
3	Germany	200	16669142023	9338645341	127.39	6.23	Drama	Harry Potter and the Chamber of S

MV 2 : Keuangan per Genre per Dekade : genre yang nge-tren, dll.

The screenshot shows a database interface with a query editor and a results table. The query is `SELECT * FROM public.mv_revenue_summary`. The results table has 10 columns: `primary_genre` (text), `decade` (integer), `total_films` (bigint), `mega_blockbuster` (bigint), `blockbuster` (bigint), `loss` (bigint), `avg_roi_pct` (numeric), `total_revenue` (numeric), and `total_profit` (numeric). The table displays 7 rows of data.

	primary_genre	decade	total_films	mega_blockbuster	blockbuster	loss	avg_roi_pct	total_revenue	total_profit
1	Action	2010	188	66	43	32	179.10	45854233587	30447602556
2	Adventure	2000	125	45	25	28	200.04	35464606380	24224115350
3	Action	2000	224	50	37	74	102.48	27386091171	15352691171
4	Comedy	2000	294	112	49	81	279.95	24723193779	16104347325
5	Drama	2000	322	102	43	118	401.28	22850141084	12930461632
6	Adventure	2010	61	17	20	10	173.35	19031318998	12291318998
7	Comedy	2010	175	71	31	34	229.22	16092294965	10495397279

MV 3 : Performa Sutradara : Sutradara paling profitable, durasi karir, dll.

The screenshot shows a database interface with a query editor and a results table. The query is `SELECT * FROM public.mv_director_performance`. The results table has 10 columns: `director` (text), `total_films` (bigint), `avg_rating` (numeric), `total_box_office` (numeric), `total_profit` (numeric), `roi_pct` (numeric), `first_film_year` (integer), `latest_film_year` (integer), and `career_years` (integer). The table displays 6 rows of data.

	director	total_films	avg_rating	total_box_office	total_profit	roi_pct	first_film_year	latest_film_year	career_years
1	David Hand	3	0.70	535692636	531066213	11478.98	1937	1942	6
2	Alex Kendrick	4	0.68	78156869	76556869	4784.80	2003	2011	9
3	Victor Fleming	3	0.70	439294426	429890426	4571.36	1939	1944	6
4	Darren Lynn Bousman	5	0.62	456342667	432154541	1800.64	2005	2016	12
5	Guy Hamilton	6	0.65	507480445	478780445	1668.22	1964	1978	15
6	Henry Joost	3	0.61	432229120	402229120	1340.76	2011	2016	6

MV 4: Ringkasan per Genre : yang paling cuan, paling diminati, tren tahunan.

The screenshot shows a database interface with a query editor and a results table. The query is `SELECT * FROM public.mv_genre_summary`. The results table has 11 columns: `genre` (text), `total_films` (bigint), `avg_rating` (numeric), `avg_popularity` (numeric), `total_revenue` (numeric), `total_budget` (numeric), `total_profit` (numeric), `avg_runtime_minutes` (numeric), `earliest_release` (integer), `latest_release` (integer), and `top_grossing_film` (text). The table displays 10 rows of data.

	genre	total_films	avg_rating	avg_popularity	total_revenue	total_budget	total_profit	avg_runtime_minutes	earliest_release	latest_release	top_grossing_film
	Action	588	6.07	34.63	91459088189	34233272285	57225815904	111.0	1930	2016	Avata
	Adventure	288	6.39	44.60	70872316207	21705338117	49166978090	116.0	1939	2016	Capta
	Drama	747	6.66	22.50	54101757502	19691630594	34410126908	121.0	1916	2016	Titani
	Comedy	634	6.06	20.32	53037118804	17399824524	35637294280	103.0	1933	2016	Shrek
	Animation	99	6.37	41.69	29595221289	8185432353	21409788936	91.0	1940	2016	Froze
	Fantasy	93	6.25	40.29	17057198167	5930438831	11126759336	108.0	1937	2016	Spide
	Science Ficti...	79	6.34	49.13	16176309703	4631307003	11545002700	113.0	1968	2016	The A

Step 4 : PySpark ETL

Concept

PySpark cocok untuk production pipeline data besar, dengan lazy execution, skala petabyte dan support spark.sql().
Distributed in-memory processing, bukan hanya pandas

Alur ETL dengan PySpark

1 EXTRACT

tabel merge_raw dari PostgreSQL via JDBC (4803 rows 22 kolom)

2 TRANSFORM 1 – Ekstraksi kolom JSON

- genres → primary_genre
- keywords → primary_keyword
- production_companies → primary_company
- production_countries → primary_country
- spoken_languages → primary_language
- crew → primary_director
- cast → lead_actor

4 TRANSFORM 3 – Agregasi per genre

```
2 FROM public.pyspark_genre_aggregation
3 ORDER BY total_revenue DESC;
```

	primary_genre text	total_films bigint	total_revenue bigint	avg_roi_pct double precision
1	Action	588	91459088189	187.51
2	Adventure	288	70872316207	471.22
3	Drama	747	54101757502	1138398.15
4	Comedy	634	53037118804	425.95
5	Animation	99	29595221289	620.67
6	Fantasy	93	17057198167	370.35
7	Science Fiction	79	16176309703	418.92

3 TRANSFORM 2 – Kalkulasi kolom baru

- profit = revenue - budget
- roi_pct = (revenue - budget) / budget × 100
- performance_label: Mega Blockbuster / Blockbuster, etc.
- popularity_tier: Viral (>100) / Popular (>50) / Known (>20) / Niche

```
2 FROM public.pyspark_tmdb_finance
3 GROUP BY performance_label
4 ORDER BY total DESC;
```

	performance_label text	total bigint
1	Mega Blockbuster	2313
2	Loss	1326
3	Profitable	629
4	Blockbuster	533
5	Break Even	2

Step 5 : Kafka + Debezium CDC

Concept

membangun sistem Change Data Capture (CDC) yang menangkap setiap perubahan data di PostgreSQL secara real-time dan mengalirkannya ke Kafka.

Step-by-Step

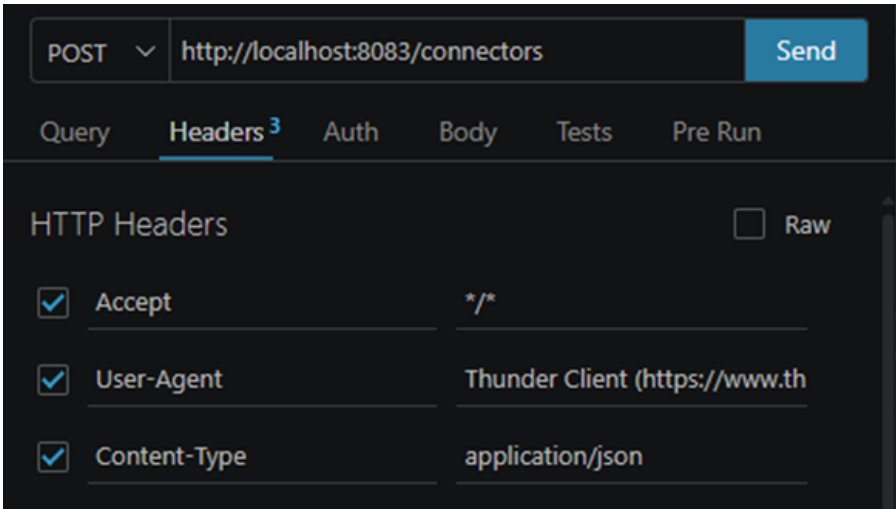
1 Komponen Stack Kafka

Container	Image	Port	Fungsi
tmdb_zookeeper	confluentinc/cp-zookeeper:7.4.0	2181	Koordinasi Kafka cluster
tmdb_kafka	confluentinc/cp-kafka:7.4.0	29092:29092	Message broker, penyimpan event stream
tmdb_debezium	debezium/connect:2.4	8083:8083	CDC connector PostgreSQL → Kafka
tmdb_kafka_man	hlebalbau/kafka-manager	9000:9000	Web UI monitoring Kafka

2 Konfigurasi PostgreSQL ke WAL Logical

3 Membuat Debezium Connector

Connector dibuat via HTTP POST ke Debezium REST API menggunakan Thunder Client:



4 Output Kafka Topics di Kafka UI

Debezium secara otomatis membuat 1 topic per tabel

Topic Kafka	Sumber Tabel
tmdb.silver_mart.dim_actor	silver_mart.dim_actor
tmdb.silver_mart.dim_company	silver_mart.dim_company
tmdb.silver_mart.dim_country	silver_mart.dim_country
tmdb.silver_mart.dim_directors	silver_mart.dim_directors
tmdb.silver_mart.dim_genres	silver_mart.dim_genres
tmdb.silver_mart.dim_language	silver_mart.dim_language
tmdb.silver_mart.dim_movies	silver_mart.dim_movies
tmdb.silver_mart.dim_revenue	silver_mart.dim_revenue
tmdb.silver_mart.dim_yearlysum	silver_mart.dim_yearlysum
tmdb.silver_mart.fact_vs	silver_mart.fact_vs
tmdb.silver_mart.fact_yearly	silver_mart.fact_yearly
+ 5 internal Kafka topics	__consumer_offsets, connect_configs, dll

(Screenshot Kafka UI)

Topic	# Partitions	# Brokers	Brokers Spread %	Brokers Skew %	Brokers Leader Skew %	# Replicas	Under Replicated %
__confluent.support.metrics	1	1	100	0	0	1	0
__consumer_offsets	50	1	100	0	0	1	0
connect-status	5	1	100	0	0	1	0
connect_configs	1	1	100	0	0	1	0
connect_offsets	25	1	100	0	0	1	0
tmdb.silver_mart.dim_actor	1	1	100	0	0	1	0
tmdb.silver_mart.dim_company	1	1	100	0	0	1	0
tmdb.silver_mart.dim_country	1	1	100	0	0	1	0
tmdb.silver_mart.dim_directors	1	1	100	0	0	1	0
tmdb.silver_mart.dim_genres	1	1	100	0	0	1	0
tmdb.silver_mart.dim_language	1	1	100	0	0	1	0
tmdb.silver_mart.dim_movies	1	1	100	0	0	1	0
tmdb.silver_mart.dim_revenue	1	1	100	0	0	1	0
tmdb.silver_mart.dim_yearlysum	1	1	100	0	0	1	0
tmdb.silver_mart.fact_vs	1	1	100	0	0	1	0
tmdb.silver_mart.fact_yearly	1	1	100	0	0	1	0

Step 6 : ClickHouse OLAP

Arsitektur 3 Layer di ClickHouse

Purpose

OLAP (column-oriented), analitik agregasi yang lebih cepat dari PostgreSQL (row oriented).)

1 BRONZE: Kafka Engine Table

- Langsung consume dari Kafka topic
- ENGINE = Kafka SETTINGS kafka_broker_list = 'tmdb_kafka:9092

2 SILVER: ReplacingMergeTree Table

- Simpan data final, hindari duplikat
- ENGINE = ReplacingMergeTree (insert_time)

3 GOLD: View

- Tambah kolom kalkulasi: profit, performance_label, popularity_tier
- WHERE budget > 0 AND revenue > 0

Output

9 Objek total di ClickHouse

Nama Objek	Tipe	Engine / Keterangan
dim_movies	Base Table	ReplacingMergeTree — data final silver layer
dim_movies_queue	Kafka Queue	Kafka Engine — consume dari topic
dim_movies_mv	Materialized View	Jembatan queue → base table (otomatis)
dim_revenue	Base Table	ReplacingMergeTree
dim_revenue_queue	Kafka Queue	Kafka Engine
dim_revenue_mv	Materialized View	Otomatis
fact_yearly	Base Table	ReplacingMergeTree
fact_yearly_queue	Kafka Queue	Kafka Engine
fact_yearly_mv	Materialized View	Otomatis

Hasil Validasi Data di ClickHouse

Tabel	Jumlah Rows	Keterangan
dim_movies	4802 rows	Data mengalir dari Kafka CDC
dim_revenue	3229 rows	Film dengan data keuangan lengkap
fact_yearly	363 rows	Agregasi per tahun per genre

```
SELECT COUNT(*)
FROM dim_revenue

Query id: 1ef6ef76-2

1. COUNT()
   3229

1 row in set. Elapse
```

```
SELECT COUNT(*)
FROM fact_yearly

Query id: 3d81fc87-

1. COUNT()
   363

1 row in set. Elapse
```

```
SELECT COUNT(*)
FROM dim_movies

Query id: 533c5984-0

1. COUNT()
   4802

1 row in set. Elapse
```


Step 7 : MinIO Object Storage

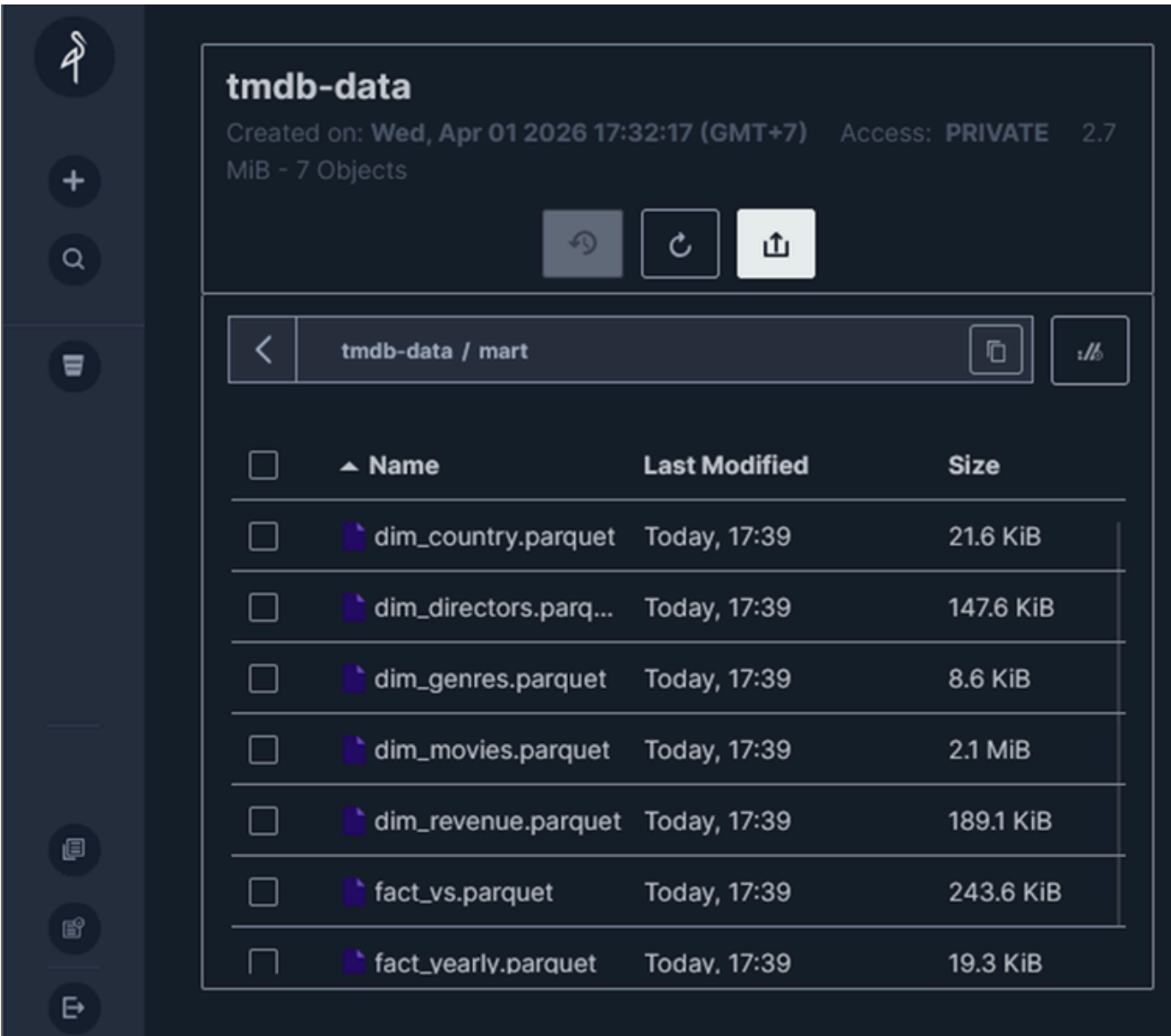
Concept

S3-compatible storage, modern pengganti HDFS. Digunakan untuk menyimpan hasil mart layer dari PostgreSQL kedalam format Parquet. Menyimpan data diluar database untuk keperluan backup, sharing atau processing lanjutan.

Output

Hasil Upload – Bucket: tmdb-data/mart
= Total: 7 file Parquet, 2.7 MiB

File Parquet	Jumlah Rows	Ukuran
dim_movies.parquet	4.802	2.1 MiB
dim_directors.parquet	2.349	147.6 KiB
fact_vs.parquet	4.867	243.6 KiB
dim_revenue.parquet	3.229	189.1 KiB
dim_country.parquet	70	21.6 KiB
fact_yearly.parquet	363	19.3 KiB
dim_genres.parquet	19	8.6 KiB



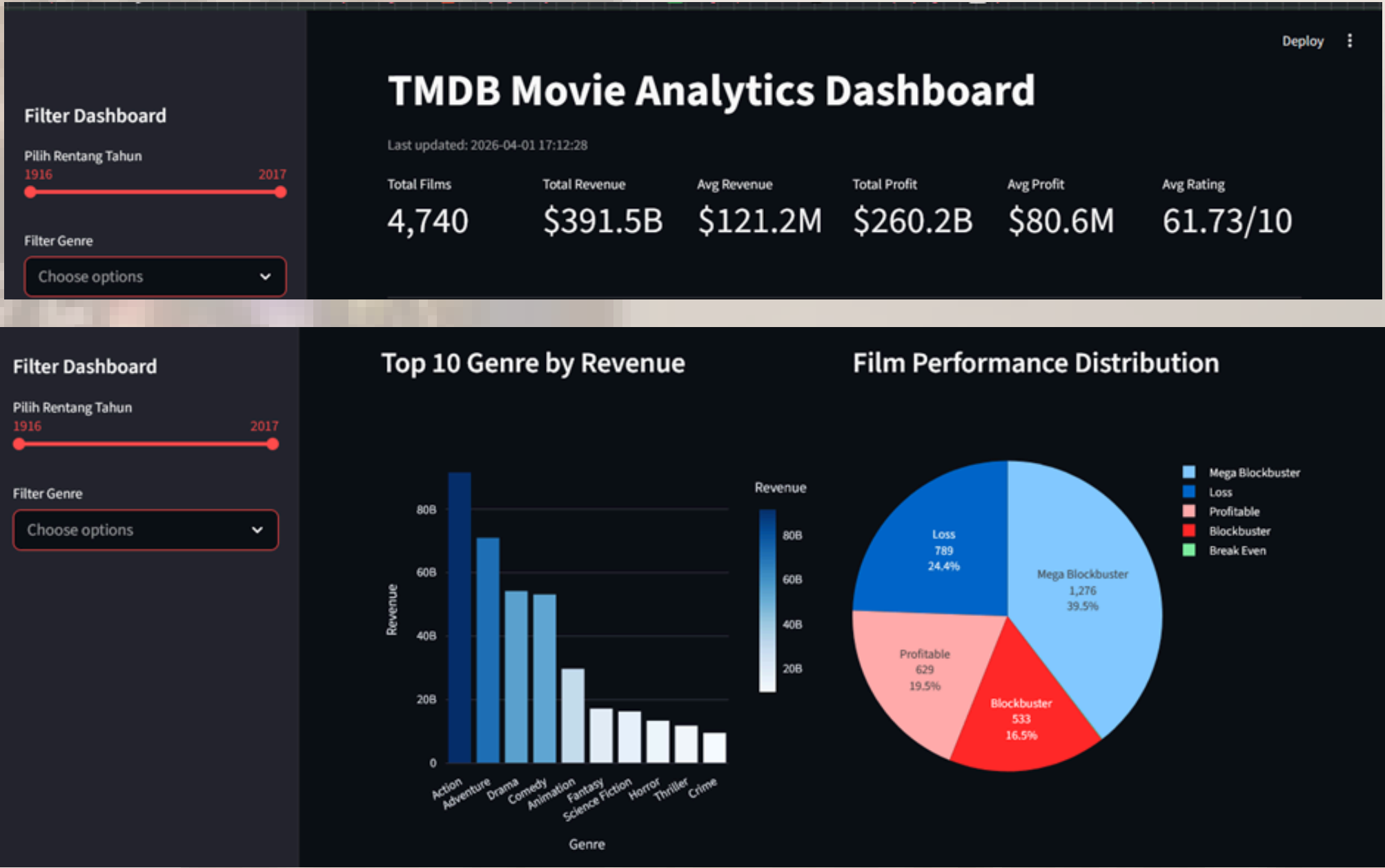
Step 8 : Streamlit Dashboard: Visualisasi Data

Purpose Dashboard interaktif dengan Streamlit (Python library) yang terhubung langsung ke data di PostgreSQL. Dashboard menampilkan scorecard KPI, chart, dan tabel dengan filter interaktif.

Fitur Dashboard

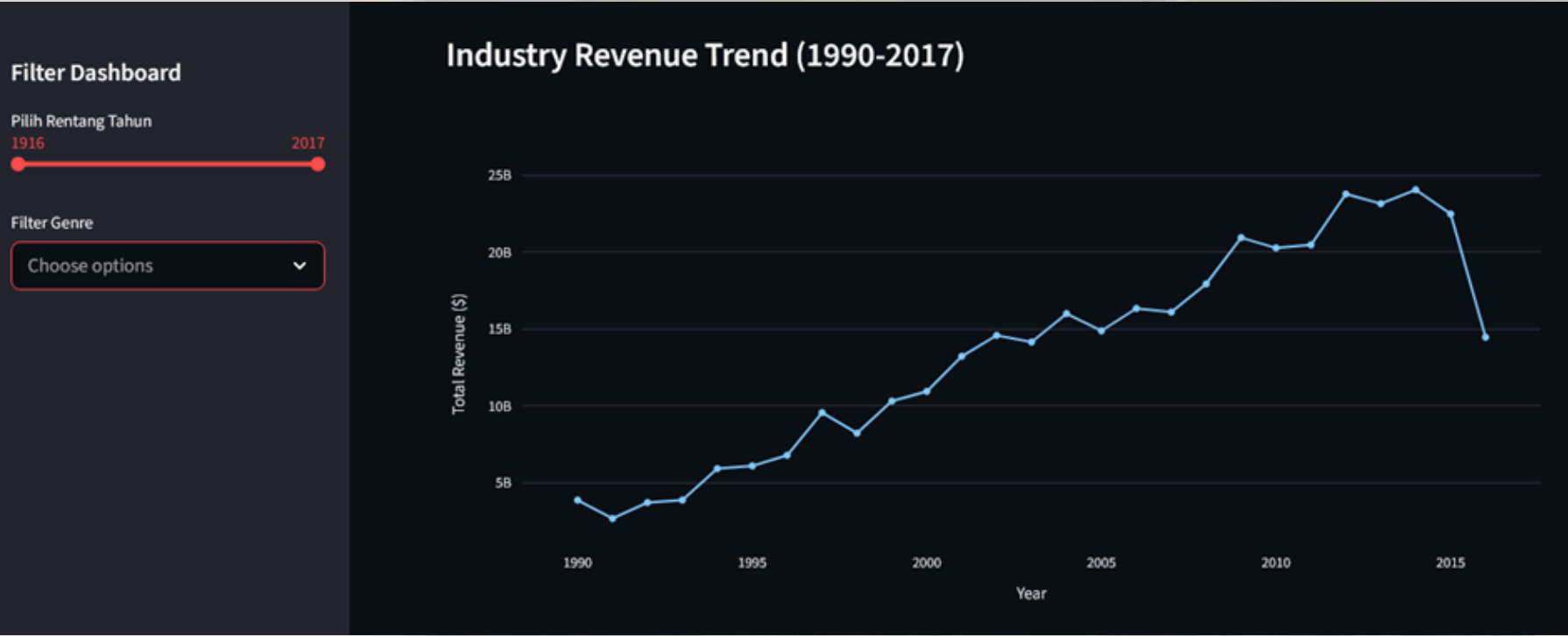
Fitur	Keterangan
Filter Rentang Tahun	Slider untuk memilih tahun rilis (misal: 1959–2004)
Filter Genre	Multi-select untuk memfilter genre (Comedy, Adventure, dll)
Scorecard KPI	Total Films, Total Revenue, Avg Revenue, Total Profit, Avg Profit, Avg Rating
Top 10 Genre by Revenue	Bar chart revenue per genre
Film Performance Distribution	Pie chart distribusi Mega Blockbuster / Blockbuster / Profitable / Loss
Top 20 Highest Revenue Films	Tabel detail dengan kolom Title, Genre, Lead Actor, Director, Revenue, Performance

Hasil Screenshot Dashboard

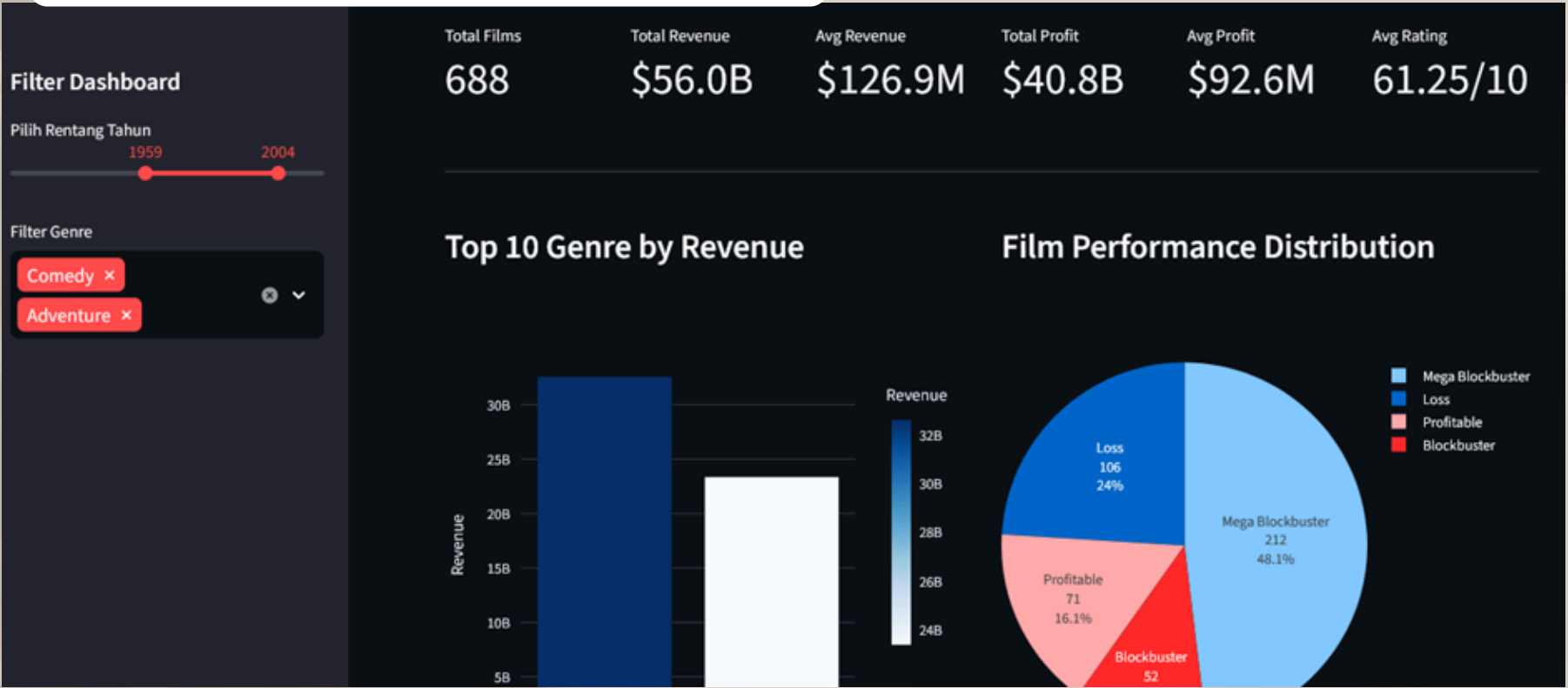


Step 8 : Streamlit Dashboard: Visualisasi Data

Hasil Screenshot Dashboard



Dashboard jika mengaktifkan filter tahun 1959-2004 dan genre Comedy dan Adventure



Filter Dashboard

Pilih Rentang Tahun
1916 2017

Filter Genre
Choose options

Top 20 Highest Revenue Films

	Title	Genre	Lead Actor	Director	Revenue	Profit	Performance	Ratin
0	Avatar	Action	Sam Worthington	James Cameron	\$2,787,965,087	\$2,550,965,087	Mega Blockbuster	7.2/1
1	Titanic	Drama	Kate Winslet	James Cameron	\$1,845,034,188	\$1,645,034,188	Mega Blockbuster	7.5/1
2	The Avengers	Science Fiction	Robert Downey Jr.	Joss Whedon	\$1,519,557,910	\$1,299,557,910	Mega Blockbuster	7.4/1
3	Jurassic World	Action	Chris Pratt	Colin Trevorrow	\$1,513,528,810	\$1,363,528,810	Mega Blockbuster	6.5/1
4	Furious 7	Action	Vin Diesel	James Wan	\$1,506,249,360	\$1,316,249,360	Mega Blockbuster	7.3/1
5	Avengers: Age of Ultron	Action	Robert Downey Jr.	Joss Whedon	\$1,405,403,694	\$1,125,403,694	Mega Blockbuster	7.3/1
6	Frozen	Animation	Kristen Bell	Chris Buck	\$1,274,219,009	\$1,124,219,009	Mega Blockbuster	7.3/1
7	Iron Man 3	Action	Robert Downey Jr.	Shane Black	\$1,215,439,994	\$1,015,439,994	Mega Blockbuster	6.8/1
8	Minions	Family	Sandra Bullock	Kyle Balda	\$1,156,730,962	\$1,082,730,962	Mega Blockbuster	6.4/1
9	Captain America: Civil Wi	Adventure	Chris Evans	Anthony Russo	\$1,153,304,495	\$903,304,495	Mega Blockbuster	7.1/1

Filter Dashboard

Pilih Rentang Tahun
1959 2004

Filter Genre
Comedy x
Adventure x

Top 20 Highest Revenue Films

	Title	Genre	Lead Actor	Director	Revenue	Profit	Performance	Ratin
0	The Lord of the Rings: The Re	Adventure	Elijah Wood	Peter Jackson	\$1,118,411,000	\$851,475,550	Mega Blockbus	7.5/10
1	Harry Potter and the Philosop	Adventure	Daniel Radcliffe	Chris Columbus	\$976,475,550	\$851,475,550	Mega Blockbus	7.5/10
2	The Lord of the Rings: The Tw	Adventure	Elijah Wood	Peter Jackson	\$926,287,400	\$847,287,400	Mega Blockbus	8.0/10
3	Star Wars: Episode I - The Pha	Adventure	Liam Neeson	George Lucas	\$924,317,558	\$809,317,558	Mega Blockbus	6.3/10
4	Jurassic Park	Adventure	Sam Neill	Steven Spielberg	\$920,100,000	\$857,100,000	Mega Blockbus	7.6/10
5	Shrek 2	Adventure	Mike Myers	Andrew Adamson	\$919,838,758	\$769,838,758	Mega Blockbus	6.7/10
6	Harry Potter and the Chambe	Adventure	Daniel Radcliffe	Chris Columbus	\$876,688,482	\$776,688,482	Mega Blockbus	7.4/10
7	The Lord of the Rings: The Fel	Adventure	Elijah Wood	Peter Jackson	\$871,368,364	\$778,368,364	Mega Blockbus	8.0/10
8	Harry Potter and the Prisoner	Adventure	Daniel Radcliffe	Alfonso Cuarón	\$789,804,554	\$659,804,554	Mega Blockbus	7.7/10
9	Star Wars	Adventure	Mark Hamill	George Lucas	\$775,398,007	\$764,398,007	Mega Blockbus	8.1/10

Step 9 : Google BigQuery: Cloud Data Warehouse

Purpose

Menyimpan 7 tabel Gold Layer dari MinIO (format Parquet) ke Google BigQuery sebagai cloud data warehouse – siap untuk analitik, BI tools, dan sharing data.

Alur-Export

- Baca 7 file Parquet dari MinIO bucket tmdb-data/mart/
- Convert ke DataFrame → Upload ke BigQuery dataset tmdb_gold
- Mode WRITE_TRUNCATE – replace data lama otomatis setiap run

Cek-Hasil:

<https://console.cloud.google.com/bigquery> → latihan-dsarea → tmdb_gold → Preview

Sukses 7/7 tabel – dijalankan via Airflow DAG task export_to_bigquery.

Screenshot BigQuery

The screenshot shows the Google BigQuery console interface. On the left, a sidebar lists the project 'latihan-dsarea' and its datasets, with 'tmdb_gold' expanded to show tables like 'dim_country'. The main panel displays the 'Schema' view for the 'dim_country' table. It includes a table with columns: Field name, Type, Mode, Description, Key, Collation, and De. The fields listed are primary_country (STRING, NULLABLE), total_films (INTEGER, NULLABLE), total_revenue (FLOAT, NULLABLE), avg_rating (FLOAT, NULLABLE), most_common_genre (STRING, NULLABLE), and most_common_language (STRING, NULLABLE). At the bottom, there are buttons for 'Edit schema' and 'View row access policies'.

Field name	Type	Mode	Description	Key	Collation	De
primary_country	STRING	NULLABLE	-	-	-	-
total_films	INTEGER	NULLABLE	-	-	-	-
total_revenue	FLOAT	NULLABLE	-	-	-	-
avg_rating	FLOAT	NULLABLE	-	-	-	-
most_common_genre	STRING	NULLABLE	-	-	-	-
most_common_language	STRING	NULLABLE	-	-	-	-

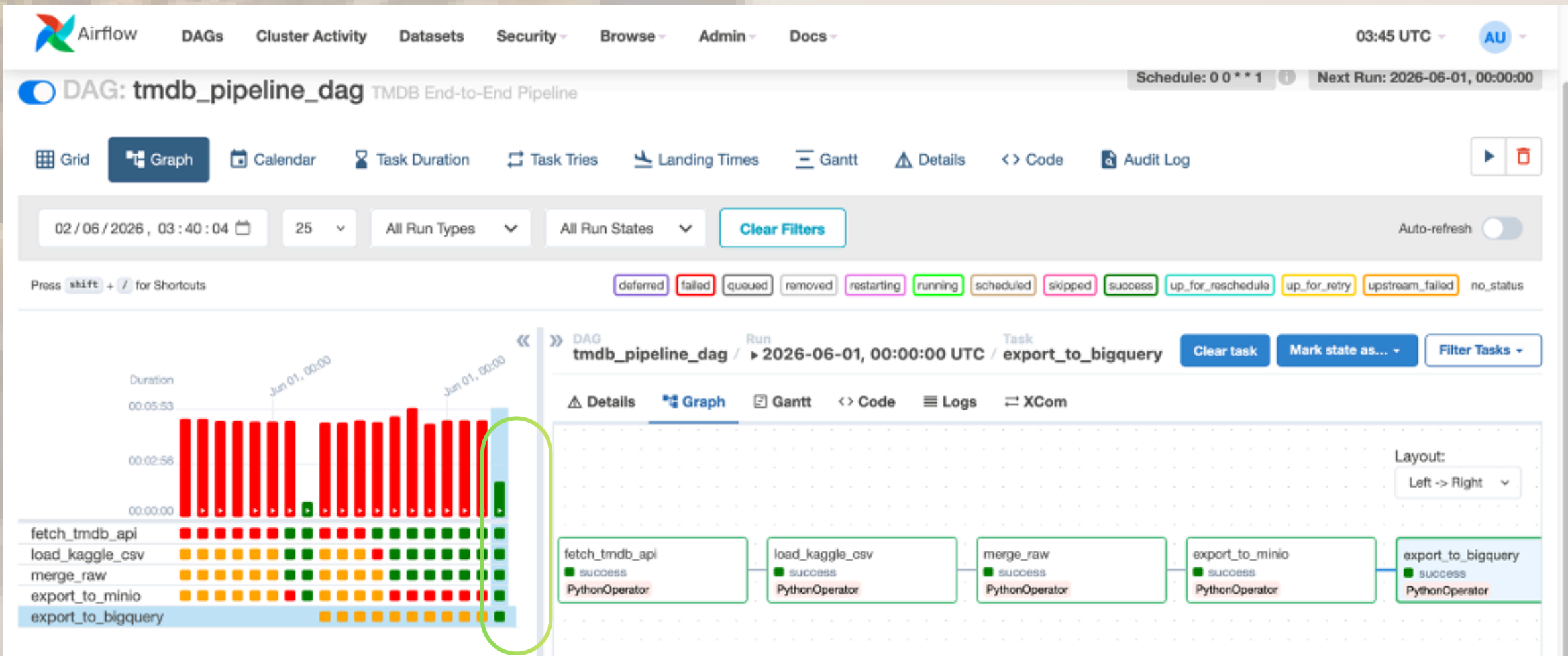
Step 10 : Apache Airflow | DAG Orchestration

Purpose Mengotomasi & menjadwalkan seluruh batch pipeline dari ingestion → dbt → PySpark → MinIO → BigQuery dalam satu DAG.

5 Task dalam DAG (Sequential)

- **Task 1 – fetch_tmdb_api :**
fetch_tmdb_api.py, load_kaggle_csv.py, merge_raw.py – ambil data dari TMDb API + load CSV Kaggle ke PostgreSQL
- **Task 2 – load_kaggle_csv :** Load file CSV Kaggle (tmdb_5000_movies + tmdb_5000_credits) ke PostgreSQL
- **Task 3 – merge_raw :** Gabungkan csv_movies_raw + csv_credits_raw menjadi tabel merge_raw (4.803 rows)
- **Task 4 – export_to_minio :** Export 7 tabel mart dari schema silver_mart_silver_mart → MinIO format Parquet
- **Task 5 – export_to_bigquery :** Baca 7 Parquet dari MinIO → Upload ke Google BigQuery dataset tmdb_gold

Screenshot Airflow DAG



KESIMPULAN

Step 1 - 8 : Step Pipeline

Step	Kategori	Nama Step	Tools / Teknologi	Output Utama
1	Data Source	Data Source	TMDB API	Raw JSON data film
2	Data Ingestion	Ingestion	Python (Requests, JSON)	Data mentah terambil
3	Streaming	Message Broker	Apache Kafka	Streaming data (topics Kafka)
4	Storage (Bronze)	Storage (Raw)	MinIO	File JSON / Parquet (raw layer)
5	Processing (Silver)	Processing	PySpark	Data transform & clean
6	Data Warehouse	Data Warehouse	ClickHouse	Tabel analytics siap query
7	Transformation	Transformation	dbt	Model dim & fact (gold layer)
8	BI / Visualization	Visualization	Streamlit	Dashboard interaktif

Capaian Proyek

1. Pipeline end-to-end tanpa orchestration tambahan Seluruh alur raw data → dashboard berjalan terintegrasi. Tanpa Airflow – setiap step terhubung dan menghasilkan output valid.
2. Arsitektur hybrid: batch + streaming Batch via dbt & PySpark, real-time via Kafka + Debezium CDC. yang dipakai di pipeline skala production.
3. Data siap analitik, performa baik, storage efisien Query cepat: index + partisi + MV di PostgreSQL & ClickHouse OLAP. Storage ringkas: 4.803 film tersimpan dalam 2.7 MiB format Parquet.